Laboratory report

A record of your saved genome analysis.

Report reference	b7d150ba3df24ae7b352799b44489fd8
Analysis completed	18 Oct 2026, 09:28 UTC
Reference assembly	GRCh38 / 1-based variant coordinates

Analysis summary

Disease-causing variants found, but they don't clearly match the symptoms

Disease-causing variants found, but they don't clearly match the symptoms. Sickle cell anemia (HBB): homozygous for a pathogenic variant. Proposed approach: homology-directed repair (correction). CRISPR advisability 76/100 (high). You carry one copy of a variant linked to Cystic fibrosis (CFTR); carriers usually have no symptoms.

Clinical eligibility has not been established. The editing score estimates predicted cutting activity, not successful correction, delivery, safety, or treatment benefit. Off-target activity requires an independent genome-wide assessment. Confirm variants and treatment suitability with a qualified clinical team.

HBB - Sickle cell anemia

Finding tier: possible | Inheritance: AR

homozygous for a pathogenic variant

Disease reference: OMIM:603903

NM_000518.5(HBB):c.20A>T (p.Glu7Val)

GRCh38 location (1-based)	Chr 11:5227002
Reference / alternate	T / A
Classification	Pathogenic; homozygous
ClinVar evidence	Variation 15333; 2/4 review stars
Phase / haplotypes	Unknown

Proposed edit: NM_000518.5(HBB):c.20A>T (p.Glu7Val)

Homology-directed repair (correction)

Rationale: The disease comes from losing working protein, so the variant should be corrected back to the healthy sequence rather than knocked out.

Alternative approach: Transversion A>T: prime editing is the precise no-break option.

Genome

Off-target assessment

Not assessed. Run a genome-wide off-target search before any lab use.

Predicted guide activity: 75.9/100; provisional threshold 50

Guides scored: 3; skipped: 0

Highest CRISPRon-predicted SpCas9 indel frequency among the proposed guides. It measures cutting, not the chance of a successful correction, and does not assess off-target activity. For homology-directed repair, the share of cut cells that are corrected is typically much lower.

Guide (5' to 3') / PAM	Strand	Cut position	Distance (bp)	Rule score	Activity /100
GTAACGGCAGACTTCTCCAC PAM AGG	+	5227000	1	109.0	57.7
CATGGTGCATCTGACTCCTG PAM TGG	-	5227005	3	101.0	53.0
CACAGGAGTCAGATGCACCA PAM TGG	+	5227017	15	55.0	75.9

Guide 1: GTAACGGCAGACTTCTCCAC

GC content: 55%; allele match: mutant only

`Model context (30 nt): GGCAGTAACGGCAGACTTCTCCACAGGAGT`

Variant is in the seed/PAM: cuts the mutant allele only

Model predictions: 1.model.best: 60.25; 2.model.best: 57.94; 3.model.best: 55.88; 4.model.best: 59.56; 5.model.best: 56.56; 6.model.best: 56.00

CRISPRoff energy terms (kcal/mol): rna_dna: -28.30; rna_dna_weighted: -50.02; dna_opening: 27.07; spacer_self_fold: -1.40; crisproff_score: 21.55

Guide 2: CATGGTGCATCTGACTCCTG

GC content: 55%; allele match: mutant only

`Model context (30 nt): ACACCATGGTGCATCTGACTCCTGTGGAGA`

Variant is in the seed/PAM: cuts the mutant allele only

Model predictions: 1.model.best: 53.44; 2.model.best: 49.88; 3.model.best: 51.00; 4.model.best: 56.62; 5.model.best: 53.94; 6.model.best: 53.12

CRISPRoff energy terms (kcal/mol): rna_dna: -27.90; rna_dna_weighted: -50.60; dna_opening: 27.00; spacer_self_fold: -0.10; crisproff_score: 23.50

Guide 3: CACAGGAGTCAGATGCACCA

GC content: 55%; allele match: also matches healthy allele

`Model context (30 nt): TCTCCACAGGAGTCAGATGCACCATGGTGT`

Model predictions: 1.model.best: 76.50; 2.model.best: 77.00; 3.model.best: 77.12; 4.model.best: 75.00; 5.model.best: 70.25; 6.model.best: 79.75

CRISPRoff energy terms (kcal/mol): rna_dna: -29.25; rna_dna_weighted: -52.71; dna_opening: 27.56; spacer_self_fold: -0.70; crisproff_score: 24.45

Genome

Repair template (121 nt)

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TCACCACCAACTTCATCCACGTTCACCTTGCCCCACAGGGCAGTAACGGCAGACTTCTCC
TCAGGAGTCAGATGCACCATGGTGTCTGTTTGAGGTTGCTAGTGAACACAGTTGTGTCAG
A
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Correcting the variant also destroys the top guide's target, preventing re-cutting.

CFTR - Cystic fibrosis

Finding tier: carrier | Inheritance: AR

one pathogenic copy (carrier)

Disease reference: OMIM:219700

NM_000492.3(CFTR):c.1521_1523del (p.Phe508del)

GRCh38 location (1-based)	Chr 7:117559590
Reference / alternate	ATCT / A
Classification	Pathogenic; heterozygous
ClinVar evidence	Variation 7105; 4/4 review stars
Phase / haplotypes	Unknown

Carrier finding: no editing target is proposed by this analysis.

Analysis record

Records scanned	2
Variants checked	2
Panel genes screened	ACADM, ALDOB, ASPA, ATP7B, BCKDHA, CFTR, F8, F9, FBN1, FGFR3, G6PD, GAA, GBA1, GLA, HBB, HEXA, HFE, MECP2, MEFV, MYBPC3, NF1, PAH, SERPINA1, TTR
Genome SHA-256	080c7a8c6d555990cb4416f8a1edd0b764ebf14073e7cdb2790ab9ad1685f91b

Research limitations

Research prototype using simulated genomes. Not a medical diagnosis or treatment recommendation; findings must be confirmed by clinical testing and a genetic counselor.